

AI models and reference designs, and strong developer productivity through the Developer Zone, alongside partnerships with AI framework vendors and edge AI ISVs. This approach ensures that customers can move from concept to product quickly and reliably.

What do you think about partnerships versus building in-house capabilities?

We develop core video, perception, and real-time inference capabilities in-house, while partnerships with framework providers and ISVs expand model ecosystems and agentic workflows. For strategic technologies, such as proprietary radar processing, we have pursued acquisitions, including Oculii, to integrate critical capabilities directly into our stack.

What is your global go-to-market strategy?

We engage directly with OEMs and Tier-1 partners, extend our reach through distributors and ecosystem partners, and enable independent developers and startups via the Developer Zone. Systems integrators contribute vertical application expertise, complementing our silicon and software capabilities.

How important is India as a market for you?

India is strategically important due to growing adoption of edge AI in smart cities and public-safety systems, a rapidly expanding automotive and EV market, and strong government support for electronics manufacturing and innovation. **India's engineering talent pool and startup ecosystem align closely with our edge-focused silicon and software strategy.**

Beyond the listed wins, what additional India partners and certifications exist, and what is the expansion timeline?

We partner with VVDN, Sparsh

(first STQC-certified CCTV range), Honeywell, eInfochips (an Arrow-acquired design house with an India presence), and eCon Systems (robotics modules showcased at CES). While we do not yet have a direct office, **we aim to establish a GTM/R&D presence and 'Made for India' products by the end of 2026** through strategic engagements.

You recently announced the CV7 SoC. What are its core highlights?

CV7 is our third-generation edge AI SoC, built on a 4nm process node, enabling ultra-low-power 8K video processing with multi-sensor fusion and on-chip AI. It supports a broad range of applications, including sports and wearable cameras with AI-based on-device editing and low-light enhancement; security cameras with on-device video understanding and vision-language model (VLM) support; automotive ADAS with localised processing for features such as automatic braking and driver monitoring; and multi-camera robotics applications that stitch multiple 4K sensors or fuse lidar and radar locally.

What are the benefits for the customers promised by CV7?

Our design is algorithm-first. We optimise memory accesses, internal pipelines, and bus architecture to minimise off-chip memory usage. **This allows multi-camera stitching, encoding, and AI inference to run simultaneously without relying on generic GPUs, which repeatedly access external memory.** On-chip AI models are loaded efficiently, reducing latency and power consumption.

How do you ensure long-term ecosystem support for developers?

By exposing our software stack, SDKs, and tools through the Developer Zone, we enable developers to iterate quickly, deploy reliably, and maintain performance. With over

400 million SoCs shipped and a consistent software platform, developers gain fast time-to-market and robust deployment support.

How do you see AI evolving from cloud-dependent systems to fully local intelligent devices?

We foresee a hybrid model in which real-time decision-making happens at the edge, while metadata learning and broader pattern recognition occur through the cloud. Edge AI will increasingly rely on smaller, domain-specific models optimised for cameras, vehicles, and robots, enabling faster and more reliable real-time action.

Where do you see the biggest growth opportunities in the next 3-5 years?

Rapid growth is expected in multi-sensor fusion platforms that combine cameras, radar, lidar, and IMUs; transformer-based and multi-modal AI models running at the edge; and embedded GenAI applications where local inference and data privacy are critical. Automotive ADAS platforms with strong vision and perception stacks are also gaining momentum, driven by the need for local intelligence, reduced cloud dependence, and real-time responsiveness.

What would be your final advice for design engineers considering Ambarella for their next product?

Understand the system-level use case, including sensor choices, AI requirements, and power constraints, and engage with our tools and ecosystem early. Early profiling and integration lead to more predictable performance and smoother field deployment. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/efymarint2>

